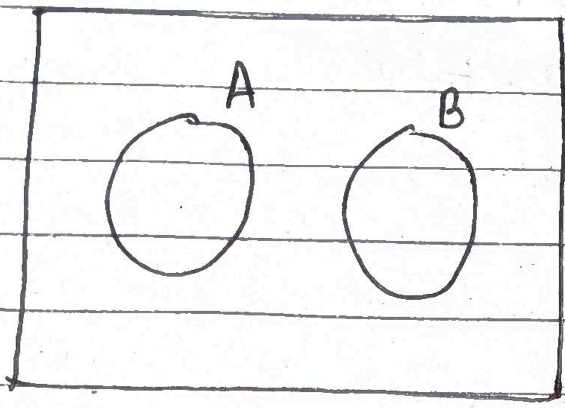
# Exam: Chapter 1 to 4

(1)

→ Sol$^n$.

For mutually exclusive events A and B,




$A \cap B = \phi$ or $AB = \phi$

$\therefore P(AB) = 0 = P(A \cap B)$

(a) $P[A|B] = P[A]$

we have,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{P(B)} = 0 \neq P[A]$$ thus the statement $P[A|B] = P[A]$ is false.

(b) $P[A \cup B|C] = P[A|C] + P[B|C]$

we have,

$$P[A \cup B|C] = \frac{P((A \cup B)C)}{P(C)} = \frac{P(AC \cup BC)}{P(C)} = \frac{P(AC)}{P(C)} + \frac{P(BC)}{P(C)} - \frac{P(ABC)}{P(C)}$$

$\Rightarrow P(A \cup B|C) = P(A|C) + P(B|C) - P(AB|C)$

for mutually exclusive, $P(A \cap B \cap C) = 0 \Rightarrow P(AB|C) = 0$

thus the statement is true.

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(c) ~~At~~ $P[A]=0$ , $P[B]=0$, or both

Mutually exclusive events implies that, the occurance of one event precludes the occurence of the other events. This doesn't imply their individual occurance probability to be zero.
Thus the given statement are False.

(d) $\frac{P[A|B]}{P[B]} = \frac{P[B|A]}{P[A]}$

LHS.

$$P[A|B] = \frac{P[AB]}{P(B)} = \frac{0}{P(B)} = 0$$

$$\therefore \frac{P[A|B]}{P[B]} = 0$$

RHS.

$$P[B|A] = \frac{P(BA)}{P(A)} = \frac{0}{P(B)} = 0$$

$$\therefore \frac{P(B|A)}{P(A)} = 0$$

Thus the given statement is True.

(e) ~~P[A|B]~~ $P[AB] = P[A] \cdot P[B]$

$\because$ $P[AB]=0$ and $P[A], P[B]$ can have positive values the given statement is False.

For statistically independent events,

(a) $P[A|B] = P[A]$

we have,

$$P[A|B] = \frac{P(AB)}{P(B)} = \frac{P(A)\,P(B)}{P(B)} = P(A)$$

This statement is True as in case of statistically independent events, the occurrence of one doesn't depend on the occurrence or non-occurrence of another event.

(b) $P[A \cup B|C] = P[A|C] + P[B|C]$

$$P(A \cup B|C) = P(A|C) + P(B|C) - P(AB|C)$$
$$= P(A) + P(B) - P(AB)$$
$$= P(A) + P(B) - P(A).P(B)$$

Since, $P(A), P(B)$ can both have positive non zero values, the given statement is False.

(c) $P[A]=0, P[B]=0$, or both

Statistically independent events doesn't imply that the probability of occurrence of individual events are zero. Thus, the given statement is False.

(d) $\frac{P[A|B]}{P[B]} = \frac{P[B|A]}{P[A]}$

LHS.

$$\frac{P[A|B]}{P[B]} = \frac{P[A]}{P[B]}$$

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RHS.

$$\frac{P[B|A]}{P[A]} = \frac{P[B]}{P[A]}$$

$\because$ LHS $\neq$ RHS. The given statement is False.

(e) $P[AB] = P[A] \cdot P[B]$

we have,

$$P[A|B] = \frac{P(AB)}{P(B)}$$

$$\Rightarrow P(A) = \frac{P(AB)}{P(B)}$$

$$\Rightarrow P(AB) = P(A) \cdot P(B)$$

Thus, the given statement is True. This is also the definition of statistically independent events.

(2)

→ Sol$^n$:

Two RC buildings A and B,
Earthquake might be strong(S), moderate (M), or weak (W),
with probabilities given as,

$P[S] = 0.02$
$P[M] = 0.20$
$P[W] = 0.78$

Let $F_A$ and $F_B$ denote the failure events of the building A and B respectively,
Then,

$P[F_A|S] = P[F_B|S] = 0.20$
$P[F_A|M] = P[F_B|M] = 0.05$
$P[F_A|W] = P[F_B|W] = 0.01$

(a) Probability of failure of building A if the impending earthquake occurs, i.e. $P[F_A]$

Using theorem of total probability,

$$P(A) = P(AE_1) + P(AE_2) + \cdots + P(AE_n)$$

Here,

$$P(F_A) = P[F_AS] + P[F_AM] + P[F_AW]$$
$$= P[F_A|S]\,P[S] + P[F_A|M]\,P[M] + P[F_A|W]\,P[W]$$
$$= 0.20 \times 0.02 + 0.05 \times 0.20 + 0.01 \times 0.78$$
$$= 0.0218$$

(b) If building A fails, the probability that the earthquake was of moderate strength, i.e. $P[M|F_A]$,

$$P[M|F_A] = \frac{P(MF_A)}{P[F_A]}$$

$$= \frac{P[F_A|M]\,P[M]}{P[F_A]}$$

$$= \frac{0.05 \times 0.20}{0.0218}$$

$$= 0.4587$$

(c)

→ given that,

$P[F_B|F_A|S] = 0.50$

$P[F_B|F_A|M] = 0.15$

$P[F_B|F_A|W] = 0.02$

we have,

$$P(E_1E_2|A) = P(E_1|E_2|A)\,P(E_2|A)$$

$$\Rightarrow P(E_1|E_2|A) = \frac{P(E_1E_2|A)}{P(E_2|A)}$$

~~P(FA AB~~

probability that both buildings A and B is failed during the earthquake, i.e. $P(F_A \cap F_B)$, we have,

$$P(F_A \cap F_B) = P(F_A \cap F_B \mid S).P(S) + P(F_A \cap F_B \mid M).P(M) + P(F_A \cap F_B \mid W).P(W)$$

Now,

$$P(F_A \cap F_B \mid S) = P(F_B \cap F_A \mid S) = P(F_B \mid F_A \mid S) \cdot P(F_A \mid S)$$
$$= 0.50 \times 0.20$$
$$= 0.1$$

Similarly,

$$P(F_A \cap F_B \mid M) = 0.15 \times 0.05 = 0.0075$$
$$P(F_A \cap F_B \mid W) = 0.02 \times 0.01 = 0.0002$$

Thus,

$$P(F_A \cap F_B) = 0.1 \times 0.02 + 0.0075 \times 0.2 + 0.0002 \times 0.78$$
$$= 0.003656$$

(d) If building A failed and B survived, the probability that the earthquake was not strong, i.e. $P[S \mid F_A \cap \overline{F_B}]$

→ we have,

$$P(F_A \cap \overline{F_B})$$
$$= P(F_A) - P(F_A \cap F_B)$$
$$= 0.0218 - 0.003656$$
$$= 0.018144$$

$$P(S \cap F_A \cap \overline{F_B}) = P(\overline{F_B} \cap F_A \cap S) = P(\overline{F_B} \mid F_A \cap S) \cdot P(F_A \cap S)$$
$$= P(\overline{F_B} \mid F_A \cap S) \cdot P(F_A \mid S) \cdot P(S)$$

we have,

$$P(\bar{E}_1 | E_2) = 1 - P(E_1 | E_2)$$

$$\Rightarrow P(\bar{F}_B | F_A \cap S) = 1 - P(F_B | F_A \cap S)$$

$$= 1 - \frac{P(F_B \cap F_A \cap S)}{P(F_A \cap S)} = 1 - \frac{P(F_B \cap F_A | S)\, P(S)}{P(F_A | S)\, P(S)}$$

$$= 1 - \frac{P(F_B \cap F_A | S)}{P(F_A | S)}$$

$$= 1 - \frac{0.1}{0.2}$$

$$= 0.5$$

$$\therefore P(S \cap F_A \cap \bar{F}_B) = P(\bar{F}_B | F_A \cap S) . P(F_A | S) . P(S)$$

$$= 0.5 \times 0.2 \times 0.02$$

$$= 0.002$$

Now,

$$P(S | F_A \cap \bar{F}_B) = \frac{P(S \cap F_A \cap \bar{F}_B)}{P(F_A \cap \bar{F}_B)} = \frac{0.002}{0.018144}$$

$$= 0.110229$$

$$\therefore P(\bar{S} | F_A \cap \bar{F}_B) = 1 - 0.110229$$

$$= 0.8898$$

(3)

→ Sol^n:

given that,

Three major loadings,

severe earthquakes (E)

loss of coolant accident (L)

thermal transients (T)

Also,

$P(E) = 0.0001$

$P(L) = 0.0002$

$P(T) = 0.00015$

$P(L|E) = 0.1$

T is independent of E and L,

(a) Probability that all types of loadings will occur, i.e. $P(E \cap L \cap T)$,

$$P(E \cap L \cap T) = P(L \cap E \cap T)$$
$$= P(T \cap E \cap L)$$
$$= P(T) \cdot P(E \cap L)$$
$$= P(T) \cdot P(L|E) \cdot P(E)$$
$$= 0.00015 \times 0.1 \times 0.0001$$
$$= 1.5 \times 10^{-9}$$

(b) The plant will shut down if any of the loadings occur,

The probability of this is, i.e. $P(E \cup L \cup T)$,

we have,

$$P(E \cup L \cup T) = 1 - P(\overline{E \cup L \cup T})$$

Using De-Morgan's law,

$$P(E \cup L \cup T) = 1 - P(\bar{E}\bar{L}\bar{T})$$
$$= 1 - P(\bar{T}\bar{E}\bar{L})$$
$$= 1 - P(\bar{T}).P(\bar{E}\bar{L})$$
$$\cancel{= 1 - P\bar{P}}$$
$$= 1 - P(\bar{T}) \cdot P(\overline{E \cup L})$$

$$= 1 - P(\bar{T}) \cdot \{1 - P(E \cup L)\}$$

$$= 1 - P(\bar{T}) \cdot \{1 - P(E) - P(L) + P(E \cap L)\}$$

$$= 1 - P(\bar{T}) \cdot \{1 - P(E) - P(L) + P(L|E).P(E)\}$$

$$= 1 - (1 - 0.00015) \cdot \{1 - 0.0001 - 0.0002 + 0.1 \times 0.0001\}$$

$$= \cancel{4.3}\ 4.4 \times 10^{-4}$$

(c) If there is a loss of revenue, the probability that severe earthquake occurred that year, i.e. $P(E|E \cup L \cup T)$

$$P(E|E \cup L \cup T) = \frac{P(E \cap (E \cup L \cup T))}{P(E \cup L \cup T)}$$

$$= \frac{P(E)}{P(E \cup L \cup T)}$$

$$= \frac{0.0001}{4.4 \times 10^{-4}}$$

$$= 0.2273$$

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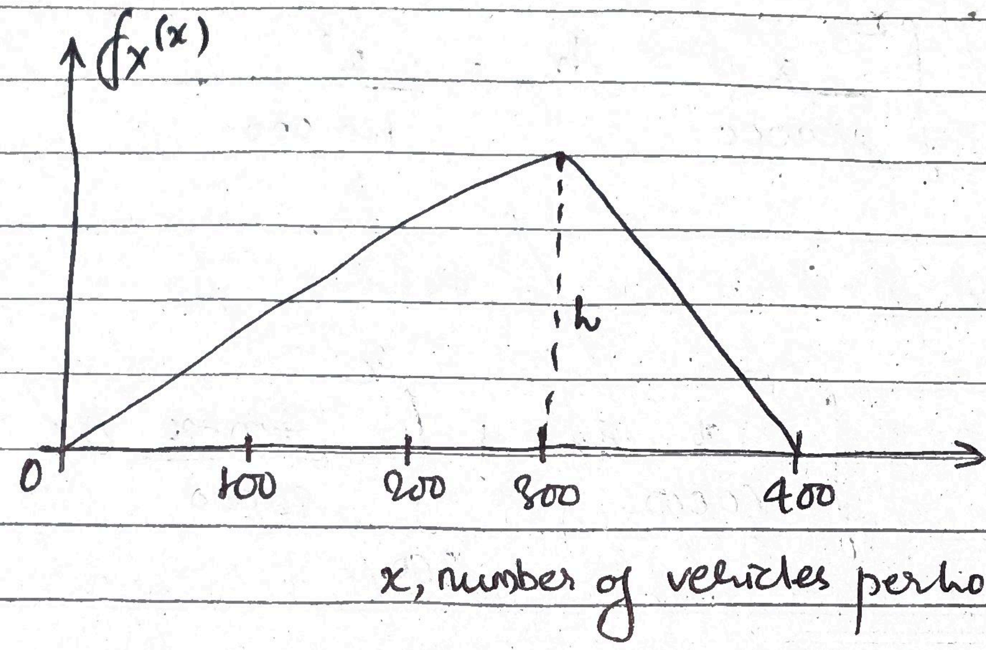



For continuous probability distributions,

$$\int f_X(x)\,dx = 1$$

$$\Rightarrow \frac{1}{2} \times 400 \times h = 1$$

$$\Rightarrow h = \frac{1}{200}$$

Thus, Probability Density Function is,

$$f_X(x) = \begin{cases} \dfrac{x}{60000}, & 0 \le x \le 300 \\ \dfrac{1}{200} - \dfrac{1}{20000}(x-300), & 300 < x \le 400 \\ 0, & \text{otherwise} \end{cases}$$

Cummulative Distribution Function,

CDF,

$$F_X(x) = P(X \le x) = \int_{-\infty}^{x} f_X(\xi)\,d\xi$$

for $0 < x \le 300$,

$$F_x = \int_0^x \frac{x}{60000}\,dx = \frac{x^2}{120\,000}$$

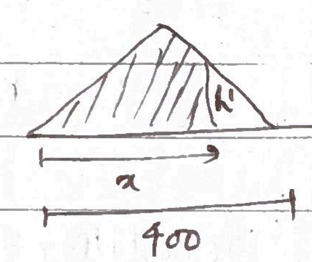
for $300 < x \le 400$,

$$F_x = \int_0^{300} \frac{x}{60000}\,dx + \int_{300}^{x} \frac{400-x}{20000}\,dx$$

$$= 0.75 + \frac{1}{200}\left[\frac{1}{50}x - \frac{x^2}{40000}\right]_{300}^{x}$$

for $300 < x \le 400$,

$$F_x = 1 - \frac{1}{2} \times (400-x)\,h'$$




$$\frac{\left(\frac{1}{200}\right)}{h'} = \frac{100}{(400-x)}$$

$$\Rightarrow h' = \frac{(400-x)}{200 \times 100}$$

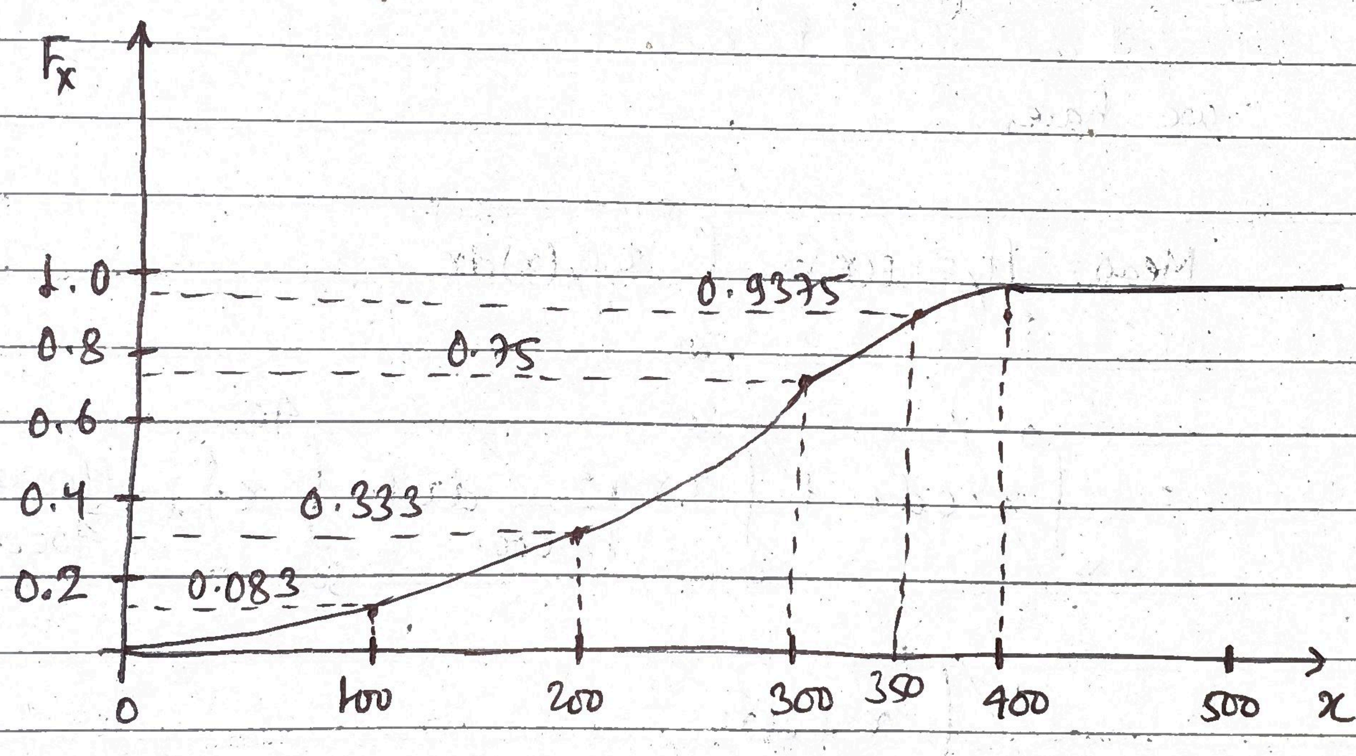
$$\therefore F_x = 1 - \frac{1}{2} \times \frac{(400-x)^2}{200 \times 100}$$

$$= 1 - \frac{(400-x)^2}{40000}$$

for $x > 400$,

$F_X = 1$

$$\therefore F_X(x) = \begin{cases} 0 & , \quad x < 0 \\ \dfrac{x^2}{120\,000} & , \quad 0 \le x \le 300 \\ 1 - \dfrac{(400-x)^2}{40\,000} & , \quad 300 < x \le 400 \\ 1 & , \quad x > 400 \end{cases} \quad \text{—— (1)}$$



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(b) Design capacity and probability of exceedance (that is, Capacity is less than traffic volume) for,

(i) The mode of X,

Mode is the value of x with largest probability density

∴ Mode = 300 vehicles per hour = Design Capacity

from eqn (1), $F_X = 0.75$

Probability of exceedance = ~~0.75~~ 1-0.75 = 0.25

(ii) The mean of X,

we have,

$$\text{Mean} = \mu_X = E(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx$$

$$= \cancel{\int_{-\infty}^{0} x \times 0\,dx + \int_{0}^{300} x \times \frac{x^2}{120000}\,dx + \int_{300}^{400} x\left\{1 - \frac{(400-x)^2}{40000}\right\}dx}$$

$$\cancel{+ \int_{400}^{\infty} x\,1\,dx}$$

$$= \int_{-\infty}^{0} x \times 0\,dx + \int_{0}^{300} x \times \frac{x}{60000}\,dx + \int_{300}^{400} x \times \frac{400-x}{20000}\,dx +$$

$$\int_{400}^{\infty} x \times 0\,dx$$

$= 0 + 150 + 83.33$ to

$= 233.33$ vph = Design Capacity

from eqⁿ (1),

$F_X = 0.4537$

Probability of exceedance = ~~0.4537~~ $1 - 0.4537 = 0.5463$

(iii) The median of X

$F_X(x_m) = 0.5$ = Probability of Exceedance (PoE)

for design capacity of 300 vph PoE = 0.75 thus design capacity lies between 0 to 300,

$$\Rightarrow 0.5 = \frac{x^2}{120000}$$

$$\Rightarrow x = 244.95 \text{ vph}$$

(iv) The mean plus one standard deviation of X,

we have,

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx$$

$$\Rightarrow E[X^2] = \int_0^{300} x^2 \times \frac{x}{60000}\,dx + \int_{300}^{400} x^2 \times \frac{400-x}{20000}\,dx$$

$$= 33750 + 27916.67 = 61666.67$$

$\therefore Var(x) = E(X^2) - \mu_x^2$

$= 61666.67 - 233.33^2$

$= 7223.78$

$\therefore$ Standard deviation, $\sigma_x = \sqrt{Var(X)}$

$\approx 84.993$

$\approx 85$

$\therefore$ Design Capacity $= 233.33 + 85 = \mu_x + \sigma_x$

$= 318.33$

from eq$^n$ ①,

$\therefore$ Probability of Exceedance $= 1 - \left[1 - \frac{(400 - 318.33)^2}{40000}\right]$

$= 1 - 0.8332 = 0.1668$

(v) the 90-percentile value, $F_X(x_{90}) = 0.9$

we have,

Probability of Exceedance $= \cancel{0.9}\ 1 - 0.9 = 0.1$

$\therefore 0.9 = 1 - \frac{(400 - \cancel{318}\, x)^2}{40000}$

$\Rightarrow x = 336.75 =$ Design Capacity

(c)

→ Actual build capacity is either 300 or 350 with relative likelihood of 1 to 4, i.e.,

$$\frac{L(300)}{L(350)} = \frac{1}{4}$$

$$\therefore \; P(X=300) = \frac{1}{1+4} = 0.2$$

$$P(X=350) = \frac{4}{1+4} = 0.8$$

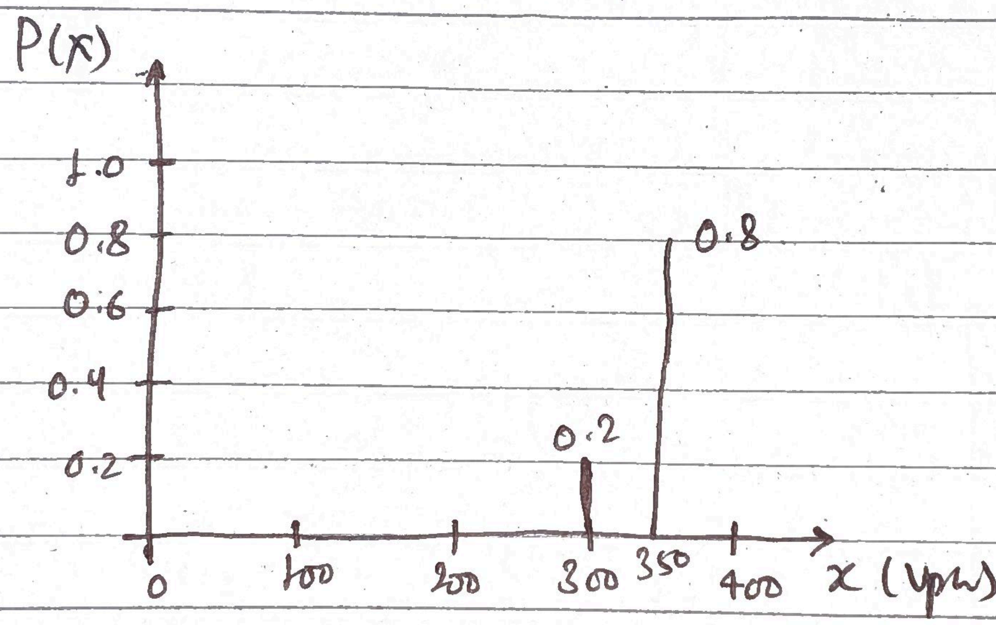


fig. Probability Mass Function

Using theorem of total probability,

probability that the capacity will be exceeded is,

$$P = 0.2 \times (\text{PoE at } X=300) + 0.8 \times (\text{PoE at } X=350)$$

$$= 0.2 \times 0.25 + 0.8 \times \left(\frac{(400-350)^2}{40000}\right)$$

$$= 0.1$$

4 ⓓ → ~~subject of chapter 9.~~

Solⁿ.

we have,

$$P''(\Theta=\theta_i) = \frac{P(\epsilon|\Theta=\theta_i)\,P'(\Theta=\theta_i)}{\sum_{i=1}^{k} P(\epsilon|\Theta=\theta_i)\,P'(\Theta=\theta_i)}$$

Where,

$P''(\Theta=\theta_i)$ = posterior probability
$P'(\Theta=\theta_i)$ = prior probability

Since the capacity of the freeway has been exceeded. The updated PMF of the

ⓐ 300 vph capacity,

$$P''(p=300) = \frac{0.2\times0.25}{0.2\times0.25+0.8\times\left(\frac{(400-350)^2}{40000}\right)}$$

$$= 0.5$$

$$P''(p=350) = \frac{\frac{(400-350)^2}{40000}}{0.2\times0.25+0.8\times\left(\frac{(400-350)^2}{40000}\right)}$$

$$= 0.5$$

(5) → Sol$^n$:

From the graph,

$$F_X(x) = \begin{cases} 0.2x &, \ 0 < x \leq 2 \\ 0.4 + 0.05(x-2) &, \ 2 < x \leq 6 \\ 0.6 + 0.1(x-6) &, \ 6 < x \leq 10 \\ 1 &, \ x > 10 \end{cases} \quad \text{——— (1)}$$

(a) probability that the progress in tunnel excavation will be between 2 and 8,

we have,

for continuous probability distribution,

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

Here,

$$P(2 < X \leq 8) = F_X(8) - F_X(2)$$
$$= 0.6 + 0.1(8-6) - 0.2 \times 2$$
$$= 0.4$$

(b) Median length ($x_m$)

we have,

$$F_X(x_m) = 0.5$$

As seen from graph $F_X = 0.5$ lies between $x = 2$ to $x = 6$,

$$\Rightarrow 0.5 = 0.4 + 0.05(x_m - 2) \quad \Rightarrow x_m = 4$$

Ⓒ Probability density function ($f_X$)

we have,

$$f_X(x) = \frac{d F_X(x)}{dx}$$

from eqⁿ ①,

$$f_X(x) = \begin{cases} 0.2 & , 0 < x \le 2 \\ 0.05 & , 2 < x \le 6 \\ 0.1 & , 6 < x \le 10 \\ 0 & , x > 10 \end{cases}$$

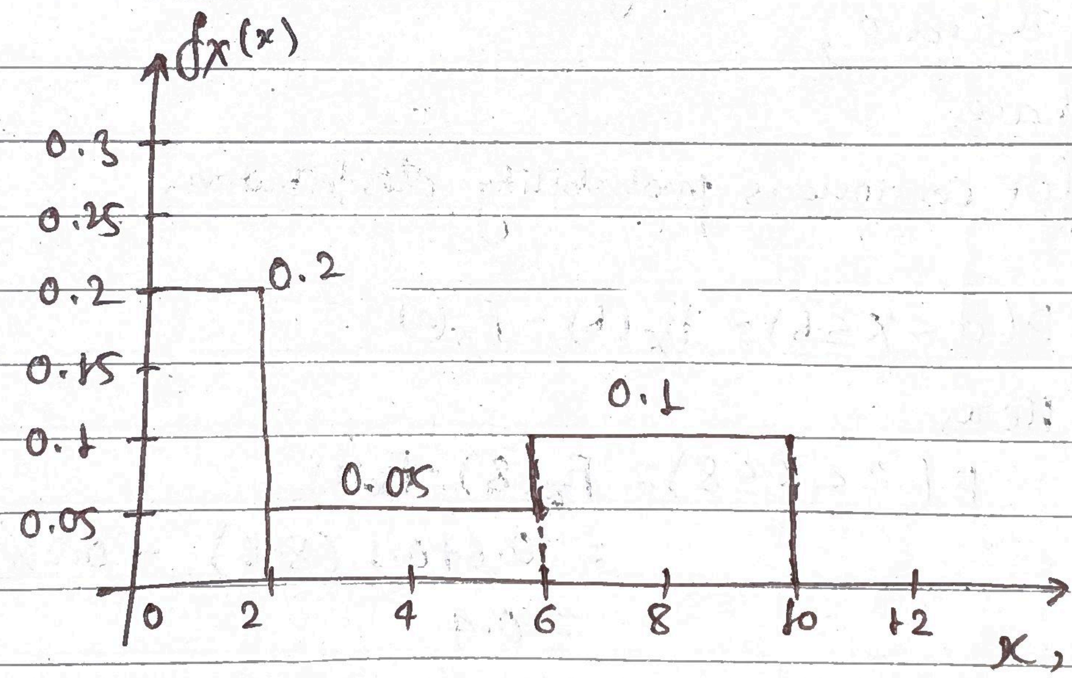


fig. ① PMF

ⓓ Mean length,

$$\mu_X = E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$= \int_0^2 \cancel{x0.2}\, x \times 0.2\, dx + \int_2^6 x \times 0.05\, dx + \int_6^{10} x \times 0.1\, dx + 0$$

$$= 0.4 + 0.8 + 3.2$$

$$= 4.4$$

(e) If the length of excavation exceeds 6m, the probability that it will exceed 8m is, i.e. $P(X>8 \mid X>6)$

we have,

$$P(X>8 \mid X>6) = \frac{P((X>8) \cap (X>6))}{P(X>6)}$$

$$= \frac{P(X>8)}{P(X>6)}$$

from fig. (1),

$$P(X>8 \mid X>6) = \frac{0.1 \times 2}{0.1 \times 4} = 0.5$$